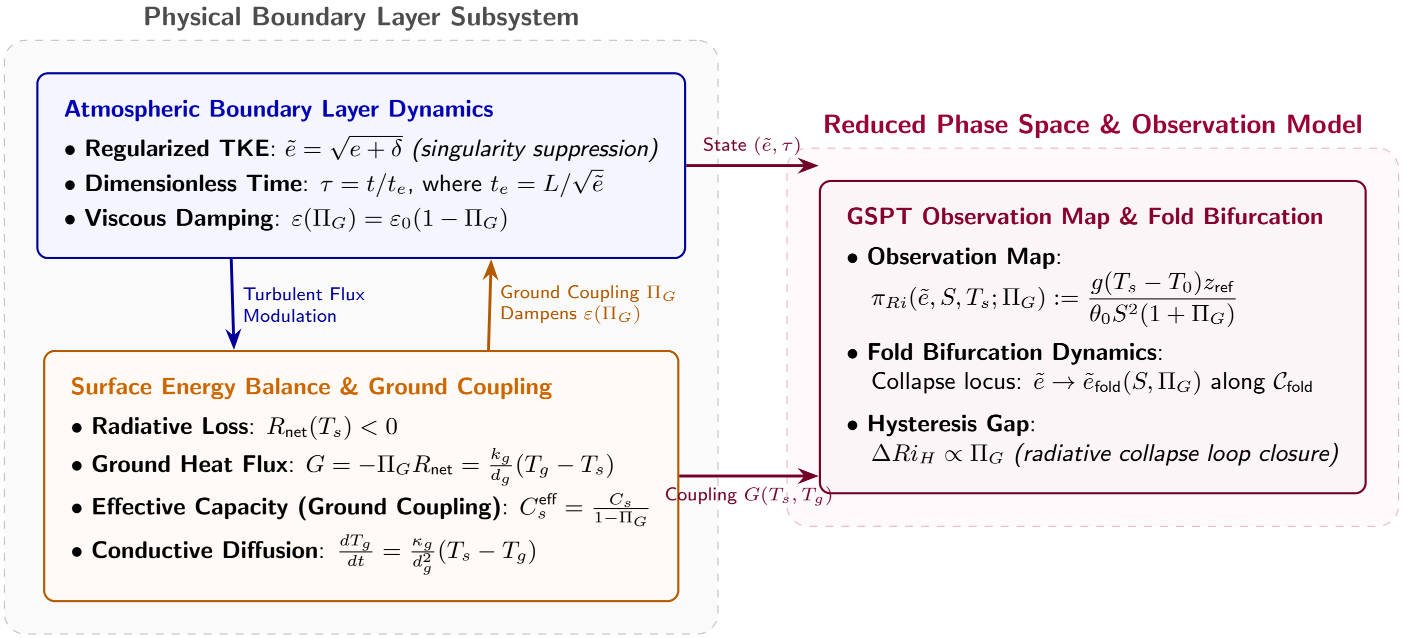
Physical Boundary Layer Subsystem
Atmospheric Boundary Layer Dynamics
• Regularized TKE: $\tilde{e} = \sqrt{e+\delta}$ (singularity suppression)
• Dimensionless Time: $\tau = t/t_e$, where $t_e = L/\sqrt{\tilde{e}}$
• Viscous Damping: $\varepsilon(\Pi_G) = \varepsilon_0(1-\Pi_G)$
Turbulent Flux Modulation
Ground Coupling Π_G Dampens $\varepsilon(\Pi_G)$
Surface Energy Balance & Ground Coupling
• Radiative Loss: $R_{\text{net}}(T_s) < 0$
• Ground Heat Flux: $G = -\Pi_G R_{\text{net}} = \frac{k_g}{d_g}(T_g - T_s)$
• Effective Capacity (Ground Coupling): $C_s^{\text{eff}} = \frac{C_s}{1-\Pi_G}$
• Conductive Diffusion: $\frac{dT_g}{dt} = \frac{\kappa_g}{d_g^2}(T_s - T_g)$
State $(\tilde{e}, \tau)$
Coupling $G(T_s, T_g)$
Reduced Phase Space & Observation Model
GSPT Observation Map & Fold Bifurcation
• Observation Map:
$\pi_{Ri}(\tilde{e}, S, T_s; \Pi_G) := \frac{g(T_s - T_0) z_{\text{ref}}}{\theta_0 S^2 (1+\Pi_G)}$
• Fold Bifurcation Dynamics:
Collapse locus: $\tilde{e} \to \tilde{e}_{\text{fold}}(S, \Pi_G)$ along $\mathcal{C}_{\text{fold}}$
• Hysteresis Gap:
$\Delta Ri_H \propto \Pi_G$ (radiative collapse loop closure)